Shuanglong Zhu & Zimo Zhang (1910 words excluding references & permission)

EV News Deconstructor

Introduction

Public debates surrounding electric vehicles (EVs) versus gasoline-powered vehicles are heavily influenced by how the media reports on environmental, economic, and technological issues. Different news sources often focus on different aspects of the same topic, leading to fragmented and even contradictory information that is difficult for readers to understand and believe. This necessitates tools that can automatically analyze news reports and link related claims to scientific facts.

Our project aims to address this problem by validating factual statements in news articles related to electric vehicles. The system takes a news article as input, automatically extracts similar factual statements, retrieves scientific evidence from a peer-reviewed corpus of scientific research on electric vehicles, assigns one of four positional labels (*SUPPORTED, NO_CONSENSUS, CONFLICTING, OUT_OF_SCOPE*), and finally generates an article-level credibility score. For example, when a user inputs an article from The Guardian, the system outputs a structured report containing extracted arguments, matching pieces of evidence, and a credibility rating from 1 to 5. Because this task requires understanding arguments in natural language and their semantic relationship to scientific text, we built our system around neural network-based semantic retrieval and natural language inference models, making natural language processing a natural fit for this problem.

Figure

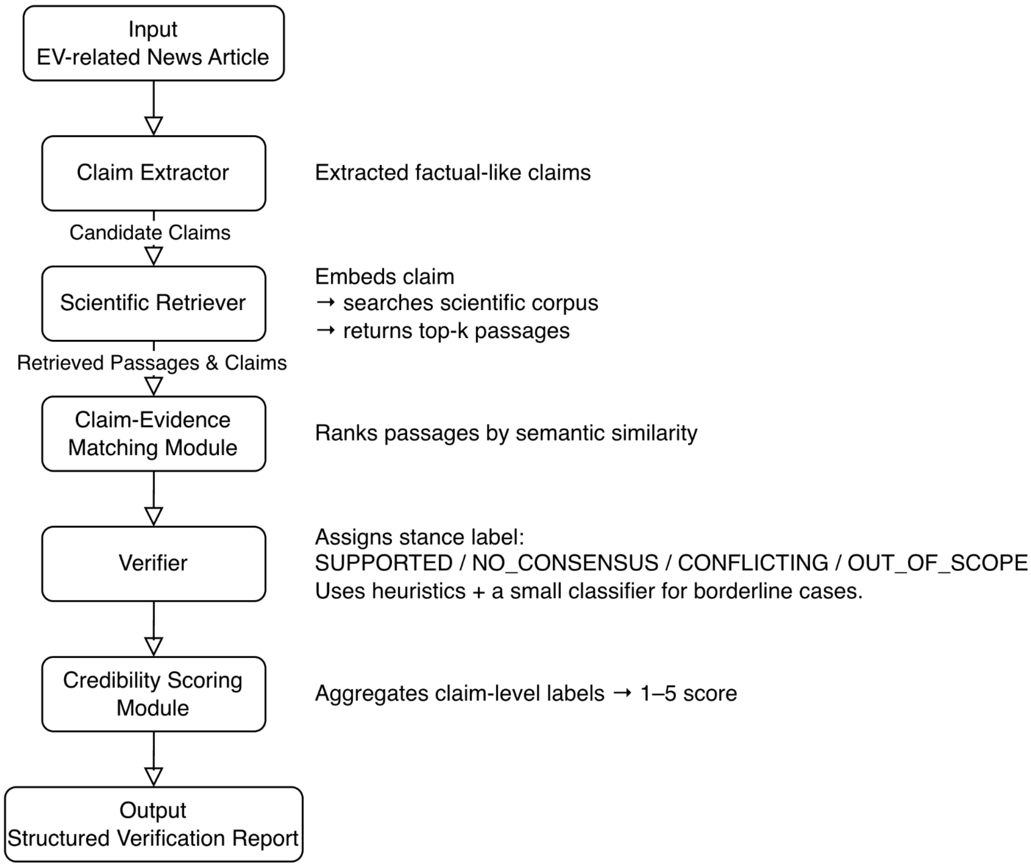

Figure 1, System Architecture

Background & Related Work

Previous research has explored using large language models to improve fact verification. Bai and Fu (2024) proposed the Full Context Retrieval and Verification (FCRV) framework, which extracts factual statements and verifies them using Retrieval-Augmented Generation (RAG). FCRV demonstrates that combining semantic retrieval with structured verification can improve the detection of fake news by retrieving external evidence and integrating it into the language model-based reasoning process. This approach directly inspired the design of our system, which also relies on retrieving a well-organized scientific corpus to evaluate the validity of news statements related to electric vehicles.

Data and Labels

Our news dataset contains 595 news articles related to electric vehicles. Due to access restrictions, 566 of these are from The Guardian, while the remaining 29 are from other media outlets and are only available through a limited free version of the NewsAPI. To support validation, we constructed a corpus of 390 scientific papers related to electric vehicles, using only their abstracts as evidence during retrieval.

Furthermore, we created a gold-standard evaluation set containing 20 manually annotated news articles from four media outlets (*BBC, The Guardian, CNN,* and *Fox*). We extracted sentences as assertions only if they expressed factual statements and could be compared to a scientific corpus. For each assertion, we assigned one of four position labels based on its fit with scientific evidence. We collectively reviewed examples and resolved disagreements through discussion to ensure consistency in annotation. The final gold-standard dataset contains 174 labeled statements distributed as follows: *SUPPORTED*(74), *NO_CONSENSUS*(75), *CONFLICTING* (1), and *OUT_OF_SCOPE*(24). The relatively small number of CONFLICTING statements reflects the nature of electric vehicle news reporting, which is characterized by a relatively small number of cases that explicitly contradict scientific findings rather than being mislabeled.

Architecture and Software

Our system follows a multi-stage pipeline combining semantic retrieval with Natural Language Inference (NLI) to verify factual claims in EV-related news. The Claim Extraction module first segments the article into sentences and identifies factual-like statements using lightweight linguistic filters, such as the presence of numerical information or EV-related keywords.

In the Semantic Retrieval stage, each claim is embedded using the BAAI/bge-small-en-v1.5 sentence-transformer model. The system initially retrieves 16 candidate evidence passages from a pre-encoded scientific corpus and filters them using three similarity thresholds (high (≥0.60), medium (≥0.50), and low (≥0.40)), retaining the top 8 most relevant sentences for verification.

The NLI Classification module uses the *cross-encoder/nli-distilroberta-base* model to evaluate the relationship between each claim and its retrieved evidence. To improve reliability, the system evaluates claim–evidence pairs using two common NLI input formulations and selects the interpretation that provides the strongest entailment or contradiction signal. The model produces probability estimates reflecting whether the evidence supports, contradicts, or is unrelated to the claim.

During Verification, these outputs are converted into the four stance labels used in our system. When multiple evidence sentences are available, their signals are combined through a weighting mechanism that accounts for both the semantic similarity of the evidence and the confidence of the NLI model, allowing the system to emphasize higher-quality evidence.

Finally, the Article Scoring component aggregates claim-level labels into a credibility rating on a 1–5 scale. The full pipeline is implemented using standard NLP and deep learning libraries, including spaCy, sentence-transformers, and HuggingFace transformers, with a lightweight web interface used to present the structured verification results.

Metrics of Success

We use two quantitative metrics to evaluate our system. Accuracy measures the proportion of claims whose predicted position labels match the human-annotated gold standard labels. The weighted-F1 accounts for label frequency by weighting F1 score of each class by its frequency in the dataset, providing a more realistic measure of performance under an imbalanced label distribution. Because some stance categories occur much more often than others, this metric provides a more representative measure of overall performance. Together, these metrics capture how reliably the system assigns stance labels to extracted claims.

Quantitative Results

Using the gold-standard set of 174 labeled claims, our system achieved an Accuracy of 0.368 and a Weighted-F1 score of 0.362. These results show that while the system can correctly assign stance labels in some cases, performance remains limited by challenges in retrieving highly relevant scientific evidence. The similarity-based retrieval stage often fails to find strong supporting passages, leading the verifier to default to more conservative labels, such as *NO_CONSENSUS*. Therefore, the quantitative results highlight retrieval quality as the primary bottleneck in improving overall stance prediction.

Qualitative Results

The way the system processes individual claims helps us understand its quantitative performance. In one article, it extracted the claim that *EVs have much lower lifecycle emissions than gasoline cars*. Although this claim is broadly supported in scientific literature, the system predicts *NO_CONSENSUS*. The retriever only returned weakly related passages that did not discuss lifecycle-emissions modeling. Without strong evidence, the verifier labeled the claim *NO_CONSENSUS*, indicating how weak retrieval can lead the system to underestimate established findings.

In contrast, the system performs well when claims highly match terminology in the scientific corpus. For example, in an article describing improvements in *EV charging performance and emission reductions*, the retriever surfaced highly relevant abstracts containing quantitative analyses of charging times and environmental impact. These abstracts matched the claims closely, allowing the verifier to assign *SUPPORTED* labels and produce a high article-level score. These examples show that system accuracy largely depends on whether retrieval returns evidence with strong semantic alignment.

Discussion and Learnings

Our results indicate that the system is functional but still limited in several important ways. The overall Accuracy (0.368) and Weighted F1 (0.362) show that while some predictions are correct, the system struggles to produce consistently reliable stance labels. A major factor is the uneven distribution of labels in the gold-standard set (*CONFLICTING* appears only once), making it effectively impossible for the model to learn this category. Although Weighted F1 provides a fairer reflection of performance by accounting for class frequency, the system still underperforms on claims that should be strongly supported. For example, the recall for *SUPPORTED* is relatively low, suggesting that the verifier often defaults to neutral judgments even when the evidence is aligned.

A notable and somewhat surprising finding is the behavior of the NLI model. Despite being well established for general natural-language inference tasks, it tends to assign very low *ENTAILMENT* probabilities on our news–scientific text pairs, even when the evidence clearly supports the claim. This contributed to misclassification and indicates a domain mismatch between the model's training data and the style of scientific abstracts. In contrast, *NO_CONSENSUS* performs better because the model frequently predicts neutral relationships, and this category appears most frequently in the dataset.

Similarity thresholds also play an important role. Raising the cutoff from 0.40 to 0.50 improves precision for *OUT_OF_SCOPE* but lowers recall substantially, demonstrating the tension between excluding irrelevant evidence and retaining useful but imperfect matches. These patterns confirm that retrieval quality is the dominant bottleneck, when relevant evidence is retrieved, the system performs well and when retrieval fails, errors cascade through NLI and verification.

Throughout development, we tested multiple design variants, but all of them showed the same pattern. We experimented with different claim-extraction heuristics, but variations in extraction did not meaningfully change downstream predictions. We also compared abstract-only retrieval with retrieval using fuller scientific text, but there still exists the problem that if retrieval failed to surface evidence that was semantically close to the claim, the verifier defaulted to neutral labels. We additionally evaluated a pre-trained verifier and a version that attempted to fine-tune the news-retrieval model on all 594 articles, but these alternatives introduced new inconsistencies without improving stance accuracy. Finally, we explored a GPT-4o end-to-end verification variant, which produced fluent explanations but showed unstable stance decisions, reinforcing the need for a structured retrieval-plus-NLI pipeline. Across all variants, retrieval quality remained the determining factor in overall performance.

Several lessons emerged from these observations. First, data quality and balance are crucial. The lack of *CONFLICTING* examples limits the system's ability to distinguish contradictory evidence, and more balanced training data would likely improve robustness. Second, retrieval quality is central to downstream performance. When claims align well with the terminology used in scientific abstracts, the system performs reliably, but retrieval failures lead to cascading errors in NLI and verification. Third, model adaptation matters. Using a generic NLI model without domain-specific tuning restricts its ability to detect scientific entailment.

In a similar project, we would spend more time understanding the data and its limitations, so we could better anticipate where the system might struggle. We would also design the pipeline around the most influential component rather than treating all parts as equally reliable. These insights would guide us in building a more robust system next time.

Fact verification using scientific literature introduces challenges not typically encountered in standard NLP tasks, because it requires robust retrieval, domain adaptation, and the coordination of multiple models. Our system incorporates these elements, reflecting the inherent complexity of the problem. Overall, the project highlights that effective fact verification depends more on evidence quality and domain-adapted reasoning than on classification alone.

Individual Contributions

Shuanglong Zhu
- Implemented most of the end-to-end codebase, including the FastAPI backend, the verification pipeline, and the integration of the BAAI/bge-small-en-v1.5 sentence-transformer and the cross-encoder/nli-distilroberta-base NLI model.
- Built and maintained the core verification modules such as semantic retrieval, claim–evidence matching, stance mapping, and article-level credibility scoring.
- Developed the evaluation scripts for computing accuracy and weighted F1 on the 174-claim gold-standard dataset, and carried out the main quantitative and qualitative error analyses.
- Contributed to the design of the system architecture figure and provided technical input for the final report and presentation.

Zimo Zhang
- Collected and organized the datasets used in the project, including 595 EV-related news articles and 390 EV-related scientific papers from Semantic Scholar.
- Led the manual annotation of the 20-article gold-standard set, including drafting the labelling guidelines and coordinating the resolution of disagreements.
- Took primary responsibility for writing and editing the project report and preparing the presentation slides, ensuring that the technical content was clearly communicated.
- Assisted with testing the end-to-end pipeline and verifying that the user interface correctly displayed claim-level and article-level verification results. and also performed some model optimizations and code-related tests.

References

[1] Y. Bai and K. Fu, “A large language model-based fake news detection framework with RAG fact-checking,” in *2024 IEEE International Conference on Big Data (BigData)*, 2024, pp. 8617–8619. doi: 10.1109/BigData62323.2024.10826000.

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